Chapter 8

Revisions and consolidation

8.1. Review strategy

Key ideas

Effective revision alternates memory recovery, explicit error correction and reuse in a problem. Passive reading of the course is not enough: you have to try a question without model, compare its answer, then repeat the question after a delay.

For each chapter, use the following code: **V** (mastered), **O** (fragile), **R** (to be continued). diagnosis, work first an orange or red skill, then immediately check with a similar question.

Skill	V	O	R	Remediation action
Choose a model	☐	☐	☐	Repeat one financial period of sorting arithmetic-tic/geometric/exponential.
Read and calculate	☐	☐	☐	Resuming targeted fluency.
Write and interpret	☐	☐	☐	Add data, formula, calculation, unit and sentence of answer.

Meeting 52 — Diagnosis of achievements

Revisions and consolidation

11/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Identify achievements.

· Sort errors.

· Plan for remediation.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient with an increase of 25 per cent. $1 + \frac{25}{100} = 1,25$
2. Give the formula of the general term arithmetic since u_0. $u_n = u_0 + nr$
3. Resolve $10^{x} = 1000$. $x = 3$
4. Derivating $x + \frac{4}{x}$. $1 - \frac{4}{x^2}$
5. State the total probability formula with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: identify the acquired and classify the errors.

Research activity

Answer without class to 14 questions, two per chapter. Report each error in a grid: knowledge, choice of method, calculation, reading, writing. Then choose two personal priorities.

Detailed correction · Professor edition

The correction awaits an error profile and a review plan, not only an overall score. verbalize the data that triggers the choice of method.

Trace written

Diagnosis is used to choose remediation: recognize error, find the useful rule, repeat a similar question, then check without help.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The correction awaits an error profile and a review plan, not only an overall score.

Training · essential

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.

2. Solve an exponential equation with a logarithm.

3. Calculate a probability from a tree and then interpret.

Detailed correction · Professor edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

Reference in the manual

Supplements: Reactivation of chapters 1 to 7. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 29, exercise 52

In France, since 2016, sales of hybrid vehicles have increased sharply. below gives the number of hybrid vehicles sold in France per year.

Year	2016	2017	2018
Number of hybrid vehicles sold	7 420	11 010	15 350

Part A

1. Determine the percentage of change between 2016 and 2017, and between 2017 and 2018.

2. Determine the geometric mean of 1.48 and 1.39.

3. What is the average annual rate of change between 2016 and 2018?

Part B

From 2018 onwards, it is assumed that hybrid car sales will increase by 43% per year and that This development will continue. (u_n) the evolution of sales where the part whole of u_n represents the number of hybrid cars sold during the year $(2018 + n)$.

1. Find u_0 ? Calculate u_1 and u_2.

2. What is the nature of the sequence? (u_n) ? Express u_n according to the n.

3. According to this model, how many hybrid vehicles will be sold in 2023?

Detailed correction · Professor edition

The rates are about 48 ,4% and 39,4 per cent The geometric mean of 1 ,48 and 1,39 value $\sqrt{1{,}48 \times 1{,}39} \approx 1{,}434$, i.e. an average of about 43 ,4 per cent in Part B, $u_0 = 15\,350$, $u_n = 15\,350 \times 1{,}43^n$; for 2023, $n = 5$ and the whole of u_5 gives about 91,788 vehicles.

Personal work

Resuming Exercise 3 with the following data:
380 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 53 — Reactivation: exponential functions

Revisions and consolidation

13/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Reactivating exponential models.

· Solve and interpret.

· Control units.

Fluency · 6 to 8 minutes, without calculator

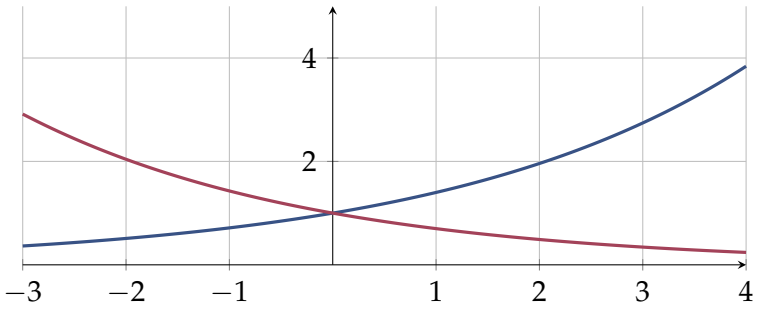
1. Reduce 3 $x + 7 - x + 2$. $2x + 9$
2. Resolve 10 $^{x} = 1000$. $x = 3$
3. Derivating $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
4. State the total probability formula with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
5. Give the formula of the general term arithmetic since u_0. $u_n = u_0 + nr$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: reactivate exponential models and resolve and interpret.

Research activity

A value of 18,000 increases by 3 ,5 % per year. Write the model, calculate the value after 8 years and determine when it will exceed 25,000.



Detailed correction · Professor edition

$V(t) = 18\,000 \times 1{,}035^{t}$, $V(8) \approx 23\,703$. The threshold is exceeded from the tenth year. verbalize the data that triggers the choice of method.

Trace written

An exponential function models a constant relative evolution. First identify the value initial and multiplier coefficient.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$V(t) = 18\,000 \times 1{,}035^t$, $V(8) \approx 23{,}703$. The threshold is exceeded from year 10.

Training · essential

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed correction · Professor edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

Reference in the manual

Supplements: Calao, pp. 48, Nos. 1-8, and then pp. 54, Nos. 47-50. is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 54, exercise 58

The number of members of a basketball club has increased in the last three years of 2 ,5 %, then 4.1% and 3.8%. Calculate the average annual rate of change.

Detailed correction · Professor edition

The overall coefficient is 1 ,025 $\times$ 1,041 $\times$ 1,038. The average annual coefficient is its cubic root, The average annual rate of change is therefore around 3.46%.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 54 — Reactivation: decimal logarithm

Revisions and consolidation

18/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Reactivate logarithms.

· Solve an equation.

· Check conditions.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 160. 8
2. Derivating $x + \frac{4}{x}$. $1 - \frac{4}{x^2}$
3. State the total probability formula with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
4. Give the formula of the general term arithmetic since u_0. $u_n = u_0 + nr$
5. Resolve 10 $^x = 1000$. $x = 3$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: reactivate logarithms and solve an equation.

Research activity

Solve 10 $^x = 340$, $1{,}06^n = 2$ and $1{,}06^n \geq 2$. For each response, distinguish actual value, value approach and minimum integer row.

Detailed correction · Professor edition

$x \approx 2{,}5315$; $n \approx 11{,}90$; for the integer threshold, $n = 12$. Verbate the data that triggers the choice of method.

Trace written

The logarithm transforms an exponential equation into an equation on the exponent. positive different from 1, second positive member.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$x \approx 2{,}5315$; $n \approx 11.90$; for the integer threshold, $n = 12$.

Training · essential

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed correction · Professor edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

Reference in the manual

Supplements: Calao, pp. 70, Nos. 1 to 10, then pp. 73 to 75, Nos. 30 to 42.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 76, exercise 43

Let x The purpose of the exercise is to determine an order of magnitude of x.

1. $x = 23.54$ and $y = 569{,}47$.
   a. The order of magnitude of a number is the power of 10 closest to that number. Give the order of magnitude of x and y.
   b. Using the calculator, determine $\log(x)$. Conjecture a link between $\log(x)$ and order of a size of x. Check your guess with y.
2. $x = 5{,}4517$. Using the calculator, determine the order of magnitude of x and $\log(x)$. The Is question 1 conjecture true?
3. Suppose that x written in scientific notation: $x = a \times 10^n$ where $a \in [1;10[$ and n is a integer giving the order of magnitude of x based on the value of a.
4. Infer a method linking the order of magnitude of a strictly positive real and His decimal logarithm.
5. Deduct the order of magnitude of the following two numbers: 2.26^{102} and $365{,}256^{2019}$.

Detailed correction · Professor edition

23,54 is closer to 10 than 100, while 569 ,47 is closer to 1,000 than 100; their orders of magnitude are 10 and 10^3. The change occurs at $a = \sqrt{10}$, not to $a = 5$. In Scientific notation $a \times 10^n$ The order of magnitude is 10^n if $a < \sqrt{10}$ and 10^{n+1} if $a \geqslant \sqrt{10}$. We applies this rule to the fractional part of logarithms: $102 \log(2{,}26) \approx 36{,}12$, so order 10^{36} ; $2019 \log(365{,}256) \approx 5173{,}88$, therefore order 10^{5174}.

Personal work

Resuming Exercise 3 with the following data:
390 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 55 — Reactivation: reciprocal function

Revisions and consolidation

20/05/2027 • B : 1 h – C : 2 h

Objectives of the session

· Reactivate reciprocal function.

· Conduct a study.

· Trace and interpret.

Fluency · 6 to 8 minutes, without calculator

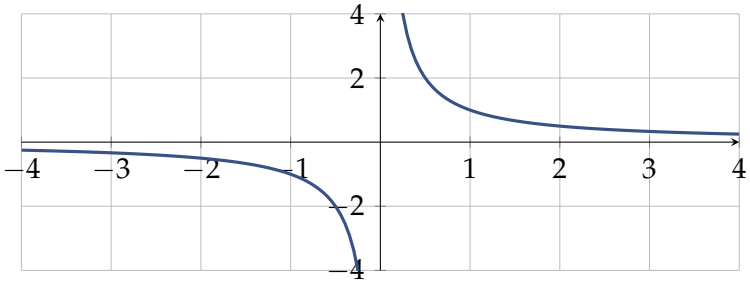
1. Write the multiplier coefficient of a 10 per cent increase. $1+\frac{10}{100}=1,10$
2. State the total probability formula with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
3. Give the formula of the general term arithmetic since u_0. $u_n = u_0 + nr$
4. Resolve $10^{x} = 1000$. $x = 3$
5. Derivating $x+\frac{4}{x}$. $1-\frac{4}{x^2}$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: reactivate the reciprocal function and conduct a study.

Research activity

Study $f(x)=x+\frac{16}{x}$ on $]0,+\infty[$ and interpret its minimum in a context of average cost. Produce the table of variations and a sketch.



Detailed correction · Professor edition

$f'(x)=\frac{(x-4)(x+4)}{x^2}$; minimum $f(4)=8$. Have the data that triggers the choice of method.

Trace written

For a function containing $\frac{1}{x}$, first exclude $x=0$, then drift, factor and study the sign before variations.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$.

Training · essential

1. Choose between arithmetic sequence, geometric sequence and exponential function in three situations.
2. Solve an exponential equation with a logarithm.
3. Calculate a probability from a tree and then interpret.

Detailed correction · Professor edition

Justify each choice by absolute or relative variation; check positivity before logarithm; multiply along a path and conclude with a sentence.

Reference in the manual

Supplements: Calao, p. 94, Nos. 1 to 9, and p. 96, Nos. 10 to 21. is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 102, exercise 47

In Murano, a glassmaker calculated that the cost of production (in euros) of x vases is given by: $C(x) = x^2 + 30x + 400$. It seeks to produce a maximum number of vases at a cost average unit as low as possible.

Part A

1. Calculate $C(0)$ and deduct the fixed costs of the craftsman. What is the production cost of 20 vases ? For 20 vases, calculate the average cost of producing a vase.
2. Note $f(x)$ the average unit cost for x vases made. For a production of at least a vase ($x \geqslant 1$), express $f(x)$ according to the x.
3. It is accepted that the average unit cost may be expressed by $f(x) = x + 30 + \dfrac{400}{x}$. Calculate $f'(x)$. Show that $f'(x) = \dfrac{(x-20)(x+20)}{x^2}$. Studying the sign of $f'(x)$ and deduce from the changes of f. How many vases must produce the craftsman in order to make the average unit cost minimal? Give that cost.

Detailed correction · Professor edition

$C(0) = 400$ € : These are fixed costs. $C(20) = 1\,400$ €, or 70 € By vase. $x > 0$, $f(x) = C(x)/x = x + 30 + 400/x$. Its derivative is $1 - 400/x^2 = ((x-20)(x+20))/x^2$. On $[1;+\infty[$, she is negative up to 20, then positive. The minimum is therefore reached for 20 vases and is worth 70 €.

Personal work

Resuming Exercise 2 with the following data:
395 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

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